The Controversial Interplay Between Hyperinflammation and Immunosenescence in Elderly Critically Ill Co-infected Patients

Wenwen SUN[2, #], Yajie Zhao[1, #], Xin Wan[3], Jie Fang[4], Zhihong XU[1, *]

[1]Department of Geriatric Medicine, Ruijin Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

[2]Shanghai Institute of Immunology, Shanghai Jiao Tong University School of Medicine, Shanghai, China

[3]Department of Information Processing Centre, Ruijin Hospital, Shanghai Jiao Tong University School of Medicine, Shagnhai, China

[4]Department of Pharmacy, Ruijin Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

[#] These authors contributed equally to this work.

*** Address correspondence to:**

Associate Prof. Zhihong Xu

Department of Geriatric, Ruijin Hospital

Shanghai Jiao Tong University School of Medicine

197 Ruijin 2nd Road, Shanghai 200025, PR China

Tel: 86-21-6445-7177

Fax: 86-21-6445-7177

Email: xzh10832@rjh.com.cn

Short running title: Immunity in Elderly Co-infection

Abstract

Background: Aging involves both immunosenescence and chronic low-grade inflammation, which can escalate into hyperinflammation during acute infection. However, how the opposing forces of immunosenescence and hyperinflammation jointly determine mortality, and whether viral co-infection exacerbates this imbalance, remains poorly understood. This study investigates these two questions in critically ill co-infected patients by examining the interplay between hyperinflammation and immunosenescence in fatal outcomes.

Methods: This retrospective cohort study enrolled critically ill co-infected patients admitted hospital from February 2023 to September 2025, following rigorous inclusion and exclusion criteria to ensure data quality. Logistic regression with elastic net LASSO regularization strategy was employed to identify risk and protective factors. Model performance was evaluated via ROC curve, calibration curve, and predicted probability distribution plot. To further understand TP-NLR for mortality prediction, the individual TP, neutrophil, and lymphocyte counts among deceased patients were examined. To investigate whether viral co-infection exacerbates the hyperinflammation-immunosenescence imbalance, clinical and laboratory variables were compared between patients with and without viral co-infection.

Results: A total of 118 critically ill patients with bacterial-fungal co-infection were enrolled (mean age 68.6 years; 83% aged ≥60 years). Multivariable logistic regression with elastic net LASSO identified total protein (TP) as an independent protective factor (OR=0.876, P=0.0006)against mortality, and neutrophil-to-lymphocyte ratio (NLR) as an independent risk factor (OR=1.023,

P=0.0218) for mortality. The combined prediction model, Logit(P)=5.830−0.132×TP+0.023×NLR, demonstrated excellent discrimination (AUC=0.841, distribution overlap=7.6%) and high calibration accuracy (MAE=0.016). Among the 23 deceased patients, 87% had profound lymphopenia, while 74% exhibited neutrophilia. Additionally, 70% patients had protein levels below 60 g/L, with 9 falling below 50 g/L. Compared to those without viral co-infection (n=39), patients with viral co-infection (n=79) exhibited a more pronounced hyperinflammation-immunosenescence imbalance, characterized by significantly higher NLR, neutrophils, PCT, and CRP, alongside lower lymphocytes, T-cell subsets, TP, and prealbumin (all FDR-adjusted P<0.05).

Conclusion: The NLR-TP model effectively predicts mortality by capturing the lethal immune imbalance in severe geriatric infections, characterized by concurrent lymphopenia, neutrophilia, and hypoproteinemia. This hyperinflammation-immunosenescence conflict is most pronounced in elderly patients with viral co-infection.

Key words: Immunosenescence; Hyperinflammation; Mortality prediction model; Critically ill; Geriatric; Co-infection

Background

Geriatric patients with severe infections represent a high-risk population, exhibiting substantially higher mortality than younger individuals [1, 2]. Aging is accompanied by profound remodeling of the immune system, termed immunosenescence, characterized by a decline in naive T cells, accumulation of memory cells, and impaired antigen presentation, leading to increased susceptibility to novel pathogens and reduced vaccine efficacy [3, 4]. Concurrently, the elderly often maintain a state of chronic low-grade inflammation known as “inflammaging”, which can escalate into uncontrolled

hyperinflammation during acute infections, contributing to tissue damage and multiple organ failure [5-7]. These seemingly contradictory processes, immunosenescence and hyperinflammation, coexist in critically ill older patients [8, 9]. However, how the opposing forces of immunosenescence and hyperinflammation jointly determine mortality remains poorly understood.

Co-infections, such as influenza complicated by bacterial and fungal pneumonia or SARS-CoV-2 co-infected with other respiratory viruses, are not uncommon in elderly patients with severe infection [10, 11]. Co-infection may further perturb immune homeostasis: persistent stimulation by multiple pathogens could accelerate T-cell exhaustion (exacerbating immunosenescence), while synergistic interactions between viruses and bacteria might provoke an even more intense inflammatory response [12]. Although co-infection has been associated with worse prognosis, direct evidence on whether it amplifies the imbalance between immunosenescence and hyperinflammation to increase mortality is lacking.

Therefore, the present study was designed to comprehensively evaluate the joint impact of immunosenescence (e.g., lymphocyte subsets, T-cell functional markers) and hyperinflammation (e.g., cytokine levels, inflammatory biomarkers) on fatal outcomes in a cohort of critically ill geriatric patients with co-infection. The primary aim was to investigate how the opposing forces of hyperinflammation and immunosenescence jointly determine mortality. The secondary aim was to assess whether the presence of viral co-infection exacerbates this immunological imbalance. Our findings may deepen the understanding of immune pathophysiology in elderly patients with severe co-infection and inform risk stratification and therapeutic strategies

Methods

Study subjects and design

This retrospective cohort study was conducted at Ruijin Hospital between February 27, 2023 and September 12, 2025. Critically ill patients with targeted next-generation sequencing (tNGS) confirmed infection were screened for eligibility. Samples included bronchoalveolar lavage fluid, sputum, blood, urine, pleural effusion, peritoneal fluid, and catheter tips.

To ensure data integrity and homogeneity of the study population, a stepwise exclusion process was applied. Patients were first excluded if lymphocyte subset testing was not performed. Further exclusions were as follow: hematologic disorders, rheumatic and autoimmune diseases, severe renal disease, solid tumors, history of solid organ or hematopoietic stem cell transplantation, enrollment in drug trials, or major trauma. Patients with missing data were excluded from the analysis. These criteria were designed to minimize confounding from pre-existing immunosuppression or non-infectious drivers of systemic inflammation.

The study was conducted in accordance with the Declaration of Helsinki and approved by the Ruijin Hospital Ethics Committee. Written informed consent was obtained from all participants.

Study variables

Medical records were reviewed to extract patient demographics, diagnoses, treatments, length of stay, survival status, hematologic parameters, imaging findings, and pathogenic test results. Primary outcomes were length of hospital stay and mortality. Analyzed blood markers focused on key immune, inflammatory and nutritional parameters, including T lymphocytes, neutrophils, C-reactive protein (CRP), procalcitonin (PCT), the neutrophil-to-lymphocyte ratio (NLR), total protein(TP), pre-albumin, red blood cell (RBC) and hemoglobin (HB). For patients undergoing fungal testing, data on T-cell subsets ($CD3^+CD4^+$, $CD3^+CD8^+$) and viral infection detection were also required.

tNGS

tNGS served as a high-throughput, cost-effective diagnostic tool capable of simultaneously detecting a broad spectrum of pathogens (bacteria, fungi, and viruses), antimicrobial resistance (AMR) genes, and virulence factors. Key detectable viruses included severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), Epstein-Barr virus (EBV), cytomegalovirus (CMV), herpes simplex virus (HSV), respiratory syncytial virus A/B (RSV A/B), influenza virus A/B (Flu A/B), and parainfluenza virus 1/2/3 (PIV 1/2/3). This technique amplified target genes at first by using one or more specific primers, and then conducted high-throughput next-generation sequencing (NGS). The accuracy of tNGS significantly exceeded that of conventional microbiological tests (CMTs), and the cost of tNGS was much lower than that of metagenomic next-generation sequencing (mNGS) [13].

Statistical Analysis

Statistical analyses were performed in R. A two-tailed P-value < 0.05 was considered statistically significant. The data included continuous, categorical, and count variables.

Univariate logistic regression was used to identify independent risk or protective factors for survival and length of hospital stay. Prior to multivariable analysis, multi-collinearity was assessed among the statistically significant variables from the univariate analysis, with the variance inflation factor (VIF) serving as the primary measure. The elastic net LASSO regularization strategy was employed to balance predictive performance and model interpretability. The discriminative ability and calibration accuracy of the final models were evaluated using the Area Under the Curve (AUC), predicted probability distribution plots (parameters of interest: difference, overlap), and calibration curves (mean absolute error, MAE). To further elucidate the role of TP-NLR in mortality prediction, an analysis was conducted on the individual TP, neutrophil, and lymphocyte counts of deceased patients.

To address whether viral co-infection exacerbates the hyperinflammation-immunosenescence imbalance, baseline demographic, clinical, and laboratory variables were compared between patients with and without viral co-infection. Normality for continuous variables was assessed using the Shapiro-Wilk test. Normally distributed variables are expressed as mean±SD (standard deviation) and compared with the independent t-test. Non-normal continuous variables are summarized as median (IQR) and compared with the Mann-Whitney U test; categorical variables are presented as frequencies and compared with the Chi-square test. P values <0.05 were considered significant, and all tests were 2-tailed. P-values were adjusted for multiple comparisons using the false discovery rate (FDR, $q < 0.05$).

Results

General characteristics of enrolled patients

Following the application of strict exclusion criteria, 118 patients were analyzed, all of whom received guideline-directed antibiotic treatment. The mean age of the cohort was 68.6 years; 83% of patients were aged≥60 years, and 63.5% were male. Most (92%) had pulmonary infections; 54% were hospitalized≥14 days, and mortality was 19.5%. 39 had bacterial-fungal (double) co-infection, while 79 had bacterial-fungal-viral (triple) co-infection. .

Results of univariate logistic regression analysis for mortality prediction

In univariate logistic regression analysis, 15 of the 27 examined variables showed a significant association with mortality ($p<0.05$); odds ratios (ORs) with 95% confidence intervals (CIs) are provided in Table 1.

Variables, including $CD3^+$ T cell count, $CD3^+CD4^+$ T cell count, $CD3^+CD8^+$ T cell count,

Table 1 The univariate and multivariate logistics analysis for mortality

Univirate logistics analysis			
Variables	OR	95%CI	P-Value
CD3 absolute count (/U)	0.998	[0.996, 0.999]	0.0019
CD4 absolute count (/U)	0.995	[0.991, 0.998]	0.0014
CD8 absolute count (/U)	0.997	[0.994, 0.999]	0.0277
CRP (mg/L)	1.006	[1.000, 1.013]	0.0526
D-dimer (mg/L)	1.300	[0.690, 2.411]	0.4054
WBC count (10^9/L)	1.088	[1.027, 1.160]	0.0056
Lymphocyte count (10^9/L)	0.258	[0.087, 0.623]	0.0066
Neutrophil count (10^9/L)	1.099	[1.037, 1.174]	0.0023
RBC count (10^12/L)	0.601	[0.336, 1.032]	0.0716
HB (g/L)	0.985	[0.966, 1.003]	0.1049
Platelet count (10^9/L)	0.991	[0.985, 0.996]	0.0021
eGFR (ml/min/1.73m2)	0.979	[0.963, 0.995]	0.0121
Creatinine (umol/L)	1.004	[0.999, 1.009]	0.1314
BUN (mmol/L)	1.049	[1.001, 1.100]	0.0421
Uric acid (umol/L)	1.002	[0.998, 1.006]	0.2984
TBIL (umol/L)	1.003	[0.990, 1.012]	0.6118
TP (g/L)	0.861	[0.797, 0.919]	<0.001
Prealbumin (mg/L)	0.987	[0.979, 0.995]	0.0014
ALT (IU/L)	0.936	[0.874, 0.994]	0.0418
AST (IU/L)	1.007	[0.960, 1.049]	0.7456
PCT (ng/MI)	1.028	[0.990, 1.075]	0.1492
NLR	1.036	[1.015, 1.062]	0.0031
Patient age (year)	1.026	[0.990, 1.070]	0.1876
Viral infection	4.068	[1.28, 18.129]	0.0320
Pleural effusion	1.722	[0.679, 4.352]	0.2472
Lung lobes involved	4.614	[1.28, 30.036]	0.0479
Patient gender	0.553	[0.186, 1.467]	0.2544

Multivariates logistics analysis				
Variables	Coefficient	OR	95%CI	P-Value
Intercept	5.830	340.219	[4.944, 35801.344]	0.0090
TP (g/L)	-0.132	0.876	[0.808, 0.941]	0.0006
NLR	0.023	1.023	[1.007, 1.046]	0.0218

ORs for lung lobes involved, viral infection, WBC, neutrophils, NLR, and BUN all exceeded 1, lymphocyte count, TP, eGFR, platelet count, and prealbumin had ORs less than 1, suggesting that higher values may be associated with a decreased odds of death. Lymphocyte count showed the strongest protective effect (OR=0.258, p=0.0066). TP was the most statistically significant finding (p<0.001) and demonstrated a consistent protective effect (OR=0.861, 95% CI: 0.797-0.919).

Table 1 Key predictors of mortality. Univariate analyses for mortality and hospital stay length identified multiple associated variables. To address multicollinearity, LASSO regularization was applied to select key independent factors for each model. For mortality, total protein (TP) and the neutrophil-to-lymphocyte ratio (NLR) were selected, with TP (OR = 0.876, 95% CI: 0.808-0.941, P = 0.0006) as a protective factor and NLR (OR = 1.023, 95% CI: 1.007-1.046, P = 0.0218) as a risk factor.

indicating a potential increased mortality risk with higher values. Among these, lung lobes involved had the strongest risk estimate (OR range: 1.27-27.17), though its wide confidence interval reflects considerable uncertainty.

Multivariates logistic analysis: TP and NLR the independent factors impacting survival status

Due to serious multicollinearity (VIF>10) among the 15 significant univariate variables, an elastic-net LASSO regularization strategy was used to optimize both predictive performance and model interpretability.

Ultimately TP and NLR were selected to construct a logistic regression model (Table 1). The model

Figure 1

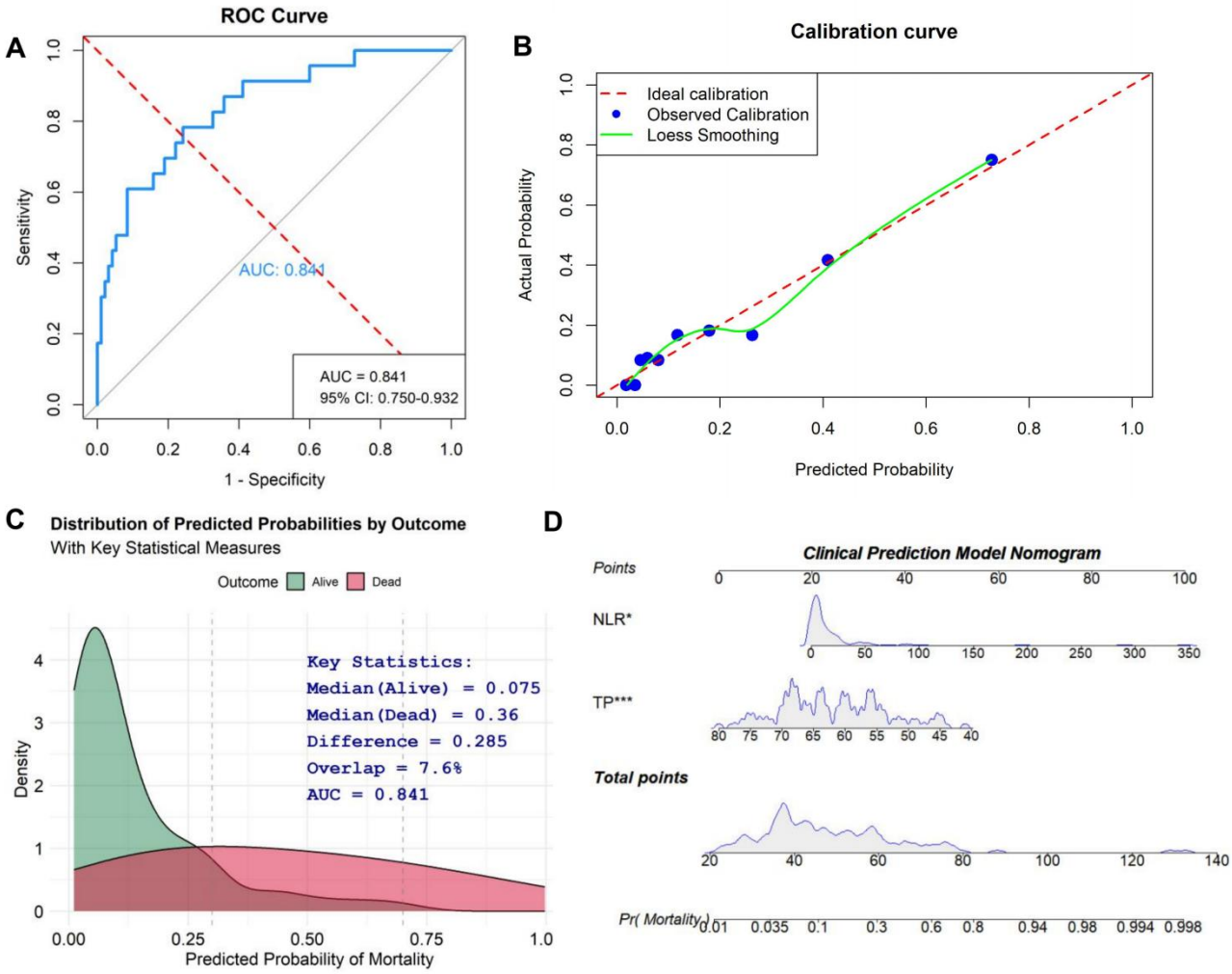

Figure 1 Performance evaluation of the mortality prediction models. (A) the ROC curve, AUC = 0.841, greater that 0.8, indicated excellent overall discriminatory ability for fatal and surviving cases. (B) The observed calibration curve close to the ideal curve (the mean absolute error < 0.05) suggested that the overall predicted probabilities were accurate and reliable. (C) In predicted probability distribution plot, the small overlap (7.6%) and substantial difference (0.285) indicated that this model exhibited high confidence in its predictions on the vast majority of samples. (D) Nomogram the intuitive graphical ruler of Logit(P1)=5.830-0.132×TP+0.023×NLR.

showed that TP has a coefficient of -0.132 (OR=0.876, 95% CI: 0.808-0.941, P=0.0006), indicating that for each one-unit increase, the odds of death decrease by approximately 12.4% on average. NLR has a coefficient of 0.023 (OR=1.023, 95% CI: 1.0071.046, P=0.0218), meaning that for each one-unit increase, the odds of death increase by approximately 2.3% on average. The final prediction model for mortality is $Logit(P_1)=5.830-0.132\times TP+0.023\times NLR$.

The model achieved an AUC of 0.841, demonstrating excellent discriminatory accuracy between deceased and surviving patients (Figure 1A). At the optimal cutoff of 0.176, accuracy, sensitivity, and specificity were 76.3%, 78.3%, and 75.8%, respectively, indicating good predictive efficacy. Calibration was excellent (MAE=0.016, <0.05), confirming reliable probability estimates (Figure 1B). The predicted probability distributions showed minimal overlap (7.6%) and a clear separation (difference=0.285), reflecting high prediction confidence for most samples (Figure 1C). The nomogram allows for direct outcome visualization (Figure 1D).

NLR was testified to be the independent risk factor for mortality. To be specific, it is worthwhile to abserve the data of neutrophil and lymphocyte counts. The table presents paired values of Neutrophil Count ($\times 10^9$/L) and Lymphocyte Count ($\times 10^9$/L).

Lymphopenia, Neutrophilia, and Hypoproteinemia in Deceased Patients

To further understand NLR as an independent risk factor for mortality, we examined the individual neutrophil and lymphocyte counts that comprise the NLR among deceased patients (Figure 2A). Specifically, 20 out of 23 deceased patients exhibited lymphocyte counts below the lowest reference value, indicating profound lymphopenia. In contrast, among these 20 patients, only one had a neutrophil count below the lowest reference value, while 17 out of the 23 patients showed neutrophil counts exceeding the upper threshold, suggesting a dominant neutrophilic response.

Total protein (TP) levels in all deceased patients were recorded at less than 70 g/L. Specifically, 16 patients fell below the normal lower limit of 60 g/L, and 9 patients had levels below 50 g/L (Figure 2B).

The simultaneous presence of lymphopenia, neutrophilia, and hypoproteinemia reflects a dysregulated immune state commonly observed in severe infections, particularly in elderly patients.

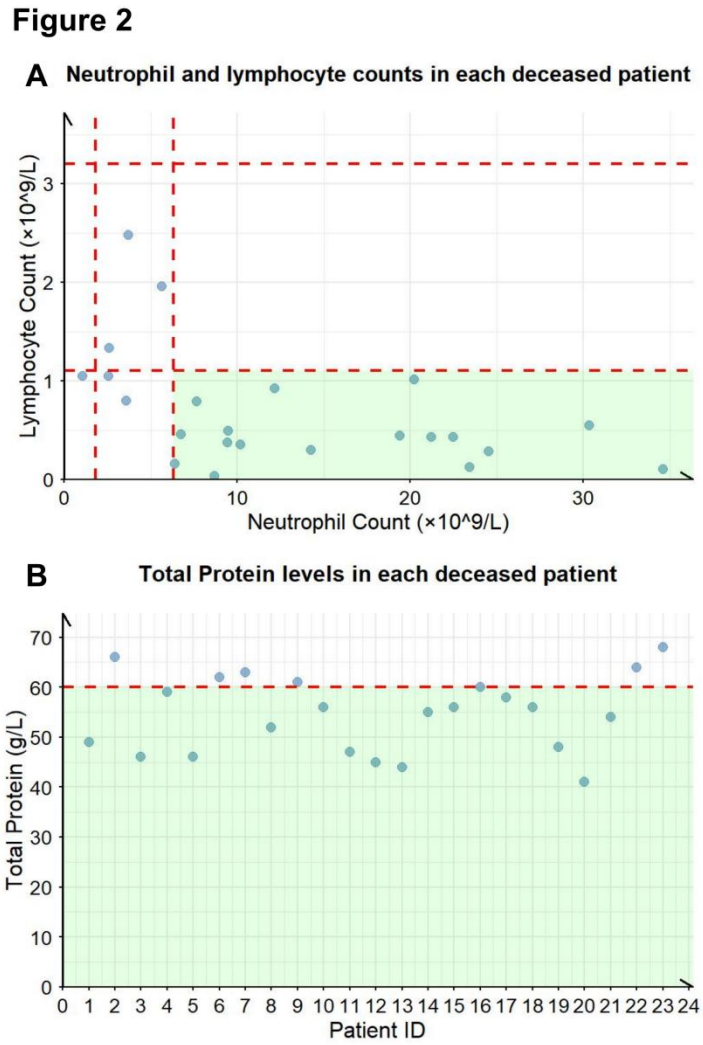

Figure 2 Levels of neutrophils, lymphocytes, and total protein in deceased patients. (A) In deceased patients, mortality was associated with a dysregulated immune state characterized by profound lymphopenia (observed in 20 of 23 patients) and a predominant neutrophilic response (present in 17 of 23 cases), rather than by low neutrophil counts. (B) Furthermore, all patients exhibited low total protein levels (<70 g/L), with the majority (16 patients) falling below the normal threshold.

Viral, Bacterial and Fungal Co-infection Exacerbates Immunological Dysregulation and Hypoproteinemia

Table 2 Statistical profile of immune dysregulation associated with polymicrobial co-infection

In univariate logistic regression analysis, viral infection was associated with increased mortality risk (OR>1). To address our secondary aim, we assessed whether viral co-infection, as a key external factor, worsened this immunological imbalance.

Non-normally distributed continuous variables: statistical description and non-parametric tests (Mann-Whitney U analysis)				
Variables	Positive(median, interquartile range)	Negative (median, interquartile range)	P-value	p.FDR(q)
CD3 absolute count (/U)	575.000 [295.000, 851.500]	906.000 [659.500, 1339.500]	<0.0001	0.0004
CD4 absolute count (/U)	248.000 [128.500, 402.500]	495.000 [315.500, 703.000]	<0.0001	0.0001
CD8 absolute count (/U)	245.000 [129.000, 372.500]	331.000 [221.000, 513.000]	0.0102	0.0215
CRP (mg/L)	54.000 [12.500, 97.300]	10.000 [5.500, 57.700]	0.0132	0.0215
D-dimer (mg/L)	1.060 [0.515, 1.640]	0.650 [0.400, 1.235]	0.0468	0.0635
WBC count (10^9/L)	9.270 [5.860, 12.685]	6.710 [5.610, 8.560]	0.0358	0.0618
Lymphocyte count (10^9/L)	0.800 [0.435, 1.240]	1.380 [0.800, 1.915]	0.0003	0.0013
Neutrophil count (10^9/L)	7.810 [4.000, 11.040]	4.510 [3.390, 6.290]	0.0032	0.0100
Platelet count (10^9/L)	180.000 [108.000, 221.000]	218.000 [162.500, 262.000]	0.0411	0.0635
eGFR (ml/min/1.73m2)	80.230 [56.700, 95.855]	85.210 [69.425, 96.850]	0.3452	0.3858
Creatinine (umol/L)	69.000 [56.000, 109.000]	71.000 [59.500, 78.500]	0.6720	0.6720
BUN (mmol/L)	8.400 [5.600, 13.350]	5.700 [4.650, 9.500]	0.0080	0.0197
Uric acid (umol/L)	226.000 [173.000, 335.000]	274.000 [203.500, 340.500]	0.0780	0.0927
TBIL (umol/L)	11.200 [8.200, 16.350]	10.100 [8.100, 14.550]	0.3940	0.4158
ALT (IU/L)	20.000 [14.000, 24.000]	22.000 [16.500, 27.000]	0.0672	0.0851
AST (IU/L)	27.000 [23.500, 34.000]	31.000 [26.000, 35.000]	0.0462	0.0635
PCT (ng/MI)	0.240 [0.090, 1.190]	0.040 [0.040, 0.175]	<0.0001	0.0004
NLR	9.680 [3.765, 20.680]	3.160 [1.850, 6.415]	<0.0001	0.0004
Patient age (year)	72.000 [65.500, 78.000]	68.000 [55.500, 72.500]	0.0083	0.0197
Normally distributed continuous variables: statistical description and t tests				
Variables	Positive (mean, sd)	Negative (mean, sd)	P-value	p.FDR(q)
RBC count (10^12/L)	3.168 (0.790)	3.773 (0.877)	0.0003	0.0005
Hemoglobin (HB) (g/L)	97.570 (23.525)	114.256 (26.637)	0.0007	0.0010
TP (g/L)	59.949 (8.342)	65.974 (7.065)	0.0002	0.0005
Prealbumin (mg/L)	143.240 (68.409)	185.205 (63.391)	0.0017	0.0017
Statistical analysis of categorical variables by using Chi-square test				
Variables	Frequency(%) Triple infection (Yes, no)	X-squared	P-value	p.FDR (q)
Patient Gender (F, M)	(32.9, 67), (43.5, 56.4)	0.866	0.3521	0.3521
Pleural effusion (Yes, No)	(43, 56.9), (25.6, 74.3)	2.676	0.1018	0.1272
Lung lobes involved (2, <2)	(82.2, 17.7), (56.4, 43.5)	7.734	0.0054	0.0135
*Statistial significance: P<0.05.				

The numerical variables RBC, HB, TP, and prealbumin were normally distributed in both groups and

Table 2 Descriptive and inferential statistics. There were lower lymphocytes ($CD3^+$ T cell, $CD3^+CD4^+$ T cells, $CD3^+CD8^+$ T cells, toal lymphocytes), nutrition signature (TP and prealbumin), RBC, HB and platelet in bacterial, fungal and viral co-infection group. However, there were higher CRP, PCT, neutrophils, NLR, and BUN in the triple co-infection group. In addition, the triple infection group showed a greater number of lobar infiltrates and longer hospital stays compared to the dual infection group. NLR: Neutrophil-to-lymphocyte ratio. HB: Hemoglobine. GFR: Estimated glomerular filtration rate. TBIL: Total bilirubin. TP: Total protein.

were therefore compared using the independent t-test (Supplementary table 1). Non-normal continuous variables and categorical variables were analyzed with the Mann-Whitney U test and Chi-square test, respectively.

Multiple key variables showed statistically significant differences after False Discovery Rate (FDR) correction (Table 2). Notably, the viral co-infection group had significantly lower levels of $CD3^+$ T cell (575.000 vs. 906.000, q=0.0004), $CD3^+CD4^+$ T cells (248.000 vs. 495.000, q = 0.0001), $CD3^+CD8^+$ T cells (245.000 vs. 331.000, q=0.0215), toal lymphocytes (0.800 vs. 1.380, q=0.0013), RBC (3.168 vs. 3.773, q=0.0005), HB (97.570 vs. 114.256, q=0.0010), TP (q=0.0005), and prealbumin (q=0.0017) . Conversely, significantly higher median values were observed for NLR (9.680 vs. 3.160, q=0.0004), neutrophils (7.810 vs. 4.510, q=0.0100), PCT (0.240 vs. 0.040, q=0.0004), and CRP (q=0.0215), and blood urea nitrogen (BUN, q=0.0197) in triple co-infection group. In addition, the triple infection group showed a greater number of lobar infiltrates and longer hospital stays compared to the dual infection group.

Discussion

This study found over 83% of patients were aged 60 or older, highlighting the high incidence of polymicrobial infections in the elderly. This aligns with existing evidence, as Hong et al. reported that co-infection risk increases with age [14]. The key finding was the development of mortality prediction models. The mortality prediction model was validated as high-quality and ready for clinical implementation.

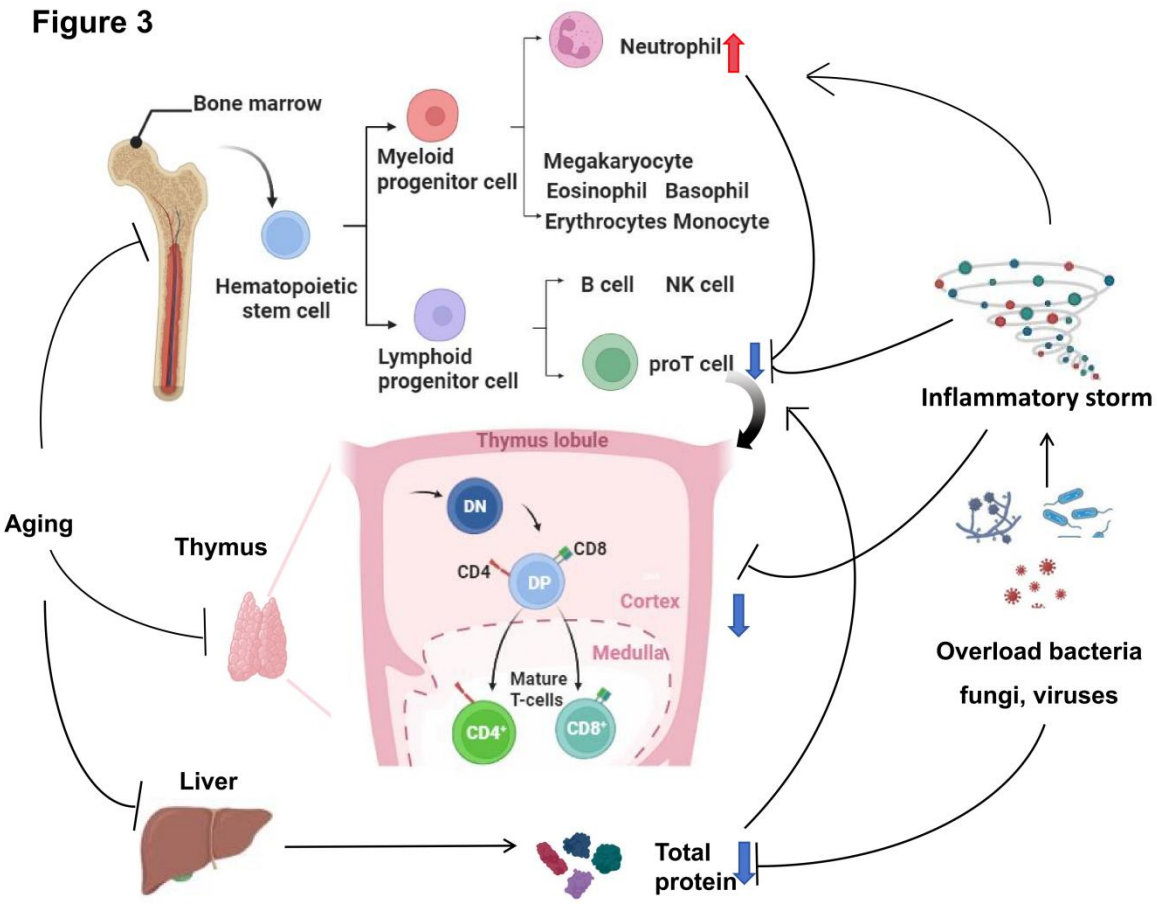

Figure 3 Mechanisms of immune dysregulation in the elderly: preserved myelopoiesis versus impaired lymphopoiesis. Neutrophils are produced, differentiated, developed, and matured entirely within the bone marrow. In contrast, T cells are produced in the bone marrow but undergo differentiation, development, and maturation in the thymus. The presence of neutrophilia in elderly patients suggests that the patient's bone marrow hematopoietic function remains relatively intact, and that inflammatory factors stimulated by pathogenic microorganisms can further promote the production of neutrophils. However, the thymus in elderly patients is atrophied; therefore, T cells, including $CD3^+CD4^+$ and $CD3^+CD8^+$ T cells, are more significantly affected by aging, leading to impaired T-cell production. Additionally, inflammatory cytokines can suppress lymphocyte generation. Furthermore, total protein (TP) not only serves as a nutrient but also participates in immune responses. Due to declining liver function in elderly patients, TP synthesis is reduced. In the context of inflammation, substantial amounts of albumin and globulins are consumed, and lymphocyte production is concurrently inhibited. These combined factors contribute to the profound decrease in lymphocytes.

Four key factors contribute to infections in the elderly: hyperinflammation, immunosenescence,

polymicrobial infection, and aging (Figure 3). In this context, NLR has been identified as an independent risk factor for mortality. Among deceased patients, the majority exhibited markedly elevated neutrophil counts (neutrophilia), accompanied by profoundly low lymphocyte (e.g., $CD3^+CD4^+$, $CD3^+CD8^+$ T cells) counts (lymphopenia). This pattern reflects the simultaneous occurrence of hyperinflammation and immunosenescence, two seemingly opposing yet coexisting immunological processes. In elderly patients facing multiple severe infections, the body enters a state of physiological emergency or stress, giving rise to the concurrent presence of both conditions [15].

In this study, hyperinflammation was characterized by increased neutrophils. Neutrophils originate from myeloid progenitor cells derived from pluripotent hematopoietic stem cells in the bone marrow [16]. As part of the innate immune system, they serve as the body’s “rapid response force” against pathogens [16]. When pathogens such as bacteria invade, the bone marrow is rapidly mobilized to produce and release neutrophils into the bloodstream [16]. Bacterial components and inflammatory factors (e.g., G-CSF, TNF-α) stimulate the bone marrow, triggering the rapid release of mature neutrophils from its reserve pool and ramping up their production [17]. The occurrence of neutrophilia suggests that bone marrow function remains preserved [17].

In the later stages of infection or in critically ill patients, the immune system may shift from an initial state of hyperactivation, characterized by neutrophilia, to a state of immune paralysis or immunosuppression [18, 19]. At this stage, lymphocyte counts decrease significantly (lymphopenia), and although neutrophil counts may remain elevated or fluctuate, their function can become impaired [19]. Patients in this state often struggle to clear pathogens, face a high risk of secondary infections, and subsequently experience increased mortality [20].

Notably, in this study, these patients exhibited lymphopenia, contrasting with the typical lymphocyte

rise seen in viral infections [21]. Immunosenescence was evidenced by lymphopenia, with a corresponding severe decline in TP (Figure 3). Immunosenescence involves weakened responses to pathogens, alongside chronic inflammation [22]. T cells are key components of the adaptive immune system. They originate from bone marrow and develop in thymus. For the geriatric, age-related thymic involution impairs T-cell production and reduces pathogen resistance [20, 21]. As summarized by Mittelbrunn et al., T-cell aging includes a shrinking TCR repertoire, naïve-memory imbalance, and cellular senescence [23].

The severe infection or hyperinflammation aggravates the reduction in lymphocyte counts, resulting in lymphopenia [24]. Three other interconnected mechanisms explain this decline. First, Excessive inflammatory factors (such as glucocorticoids and TNF-α) induce programmed cell death in lymphocytes [25]. Second, persistent antigenic stimulation drives lymphocytes into a state of exhaustion, characterized by impaired function and reduced proliferative capacity, further depleting their numbers [25]. Third, under the influence of inflammatory chemokines, lymphocytes exit the bloodstream and migrate to infected tissues or organs (e.g., the lungs) to fight pathogens, leading to a decrease in detectable peripheral blood counts [25]. This impaired response is primarily attributed to declined immune function and a heightened inflammatory state in the elderly [26].

Total protein includes albumin and globulin. Serum albumin (SA) can bind various inflammatory mediators and participates in regulating immune responses during systemic inflammation [27]. The conformational state of albumin can influence the mechanical properties of erythrocytes [28]. Globulins include antibodies, complement proteins, cytokines, and acute phase proteins [29, 30]. While TP is often dismissed as a mere nutritional marker, it is actually associated with humoral immunity, complement activity, and metabolic reserve [30]. The nutrient-sensing system is dysregulated in the

elderly [31]. Elevated inflammation and oxidants accelerate the loss of TP [32]. Hence, an excessive reduction in TP can lead to immunosenescence.

RBC count and HB concentration were significantly lower than in the viral-negative group. They function not only in oxygen transport but also in pathogen defense and innate immune regulation [33]. During infection, pathogens may acquire iron from HB to enhance virulence; conversely, bacterial proteases trigger the activation of HB, which in turn produces reactive oxygen species (ROS) that kill bacteria.[34].

In this research, when elderly patients with bacterial and fungal co-infection acquired a viral infection, a significant decrease in lymphocytes, total protein, and hemoglobin, along with an increased NLR, was observed.When different pathogens (viruses, bacteria, fungi) infect the same host cell, they can actively assist each other in disrupting epithelial barriers and evading the immune system by binding with each other [35, 36]. In addition, high-level pathogenic stimulation induces T-cells to express multiple inhibitory receptors (like PD-1, TIGIT) [37]. When a virus infects an organism burdened with senescent cells, it triggers a dangerously amplified response [38]. The virus causes an accumulation of reactive oxygen species in these senescent cells [39]. This, in turn, reduces the expression of a protein called PDLIM2, which normally acts as a brake on the powerful inflammatory pathway NF-κB [39].

This study has limitations inherent to its single-center, retrospective design. To enhance reliability, we rigorously selected subjects and employed robust statistical methods to control for imbalances. The findings would benefit from validation in multi-center, prospective studies.

Conclusion:

This study reveals a critical interplay between hyperinflammation and immunosenescence in elderly critically ill co-infected patients, quantified by the NLR and TP. While TP emerged as an independent protective factor, NLR was a significant risk factor for mortality. The combined NLR-TP model demonstrated excellent predictive performance for mortality, with high discriminative ability and calibration accuracy. Deceased patients were uniformly characterized by profound lymphopenia with neutrophilia, reflecting immune dysregulation, and severe hypoproteinemia. Viral co-infection further intensified this lethal imbalance, exacerbating both NLR elevation and TP depletion. These findings position the NLR-TP model as a clinically interpretable tool for capturing this complex host response and guiding early risk stratification. Further multicenter, prospective validation is needed to confirm the robustness and clinical applicability of the NLR-TP model. Additionally, its cut-off values and calibration should be assessed across diverse populations.

Acknowledgments

The authors greatly appreciate all the patients involved in the study.

Disclosure statement

The authors declare no conflicts of interest.

Author Contributions

All authors were involved in the study design, data acquisition, analysis, and manuscript preparation. They reviewed, approved the final manuscript, and agree to be accountable for the work.

Data availability statement

Due to ethical restrictions and participant confidentiality agreements, the data that support the findings of this study are not publicly available. They are available on reasonable request from the corresponding author, professor Zhihong Xu, subject to approval by the Ethics Committee of Shanghai Jiao Tong University Affiliated Ruijin Hospital and the execution of a data use agreement.

Funding

This work was supported by Shanghai Municipal Health Commission Research Project Funding (2025BJPT010), National Natural Science Foundation of China (82103079) and Shanghai Municipal Health Commission (202140052).

Ethics Approval and Informed Consent

This study was approved by the Ethics Committee of Ruijin Hospital (No. 2023-144) and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants.

Clinical Trial Number

Not applicable.

Reference

1. Zhu Y, Liu Y, Jiang H. Geriatric Health Care During the COVID-19 Pandemic: Managing the Health Crisis. Clin Interv Aging. 2022;17:1365-78. https://doi.org/10.2147/CIA.S376519.

2. Wu Y, Goplen NP, Sun J. Aging and respiratory viral infection: from acute morbidity to chronic sequelae. Cell Biosci. 2021;11(1):112. https://doi.org/10.1186/s13578-021-00624-2.

3. Yousefzadeh MJ, Flores RR, Zhu Y, et al. An aged immune system drives senescence and ageing of solid organs. Nature. 2021; 594(7861):100-5. https://doi.org/10.1038/s41586-021-03547-7.

4. Wang Y, Dong C, Han Y, et al. Immunosenescence, aging and successful aging. Front Immunol. 2022;13:942796. https://doi.org/10.3389/fimmu.2022.942796.

5. Bachi ALL, Abbasi A, Durigan JLQ, et al. Editorial: The role of latent chronic infection in immunosenescence and inflamm-aging. Front Immunol. 2023;14:1285234. https://doi.org/10.3389/fimmu.2023.1285234.

6. Sharma R. Exploring the emerging bidirectional association between inflamm-aging and cellular senescence in organismal aging and disease. Cell Biochem Funct. 2024;42(2):e3970. https://doi.org/10.1002/cbf.3970.

7. Choi H, Shin EC. Hyper-inflammatory responses in COVID-19 and anti-inflammatory therapeutic approaches. BMB Rep. 2022;55(1):11-9. https://doi.org/10.5483/BMBRep.2022.55.1.152.

8. Teissier T, Boulanger E, Cox LS. Interconnections between Inflammageing and Immunosenescence during Ageing. Cells. 2022;11(3):359. https://doi.org/10.3390/cells11030359.

9. Wang S, Huo T, Lu M, .et al. Recent Advances in Aging and Immunosenescence: Mechanisms and Therapeutic Strategies. Cells. 2025;14(7):499. https://doi.org/10.3390/cells14070499.

10. WHO. (October 31, 2025). COVID-19. https://data.who.int/dashboards/covid19/cases?n=c

11. Huang HY, Lin YC, Lu PL, et al. Clinical insights into mixed Candida and bacterial bloodstream infections: a retrospective cohort study. Microbiol Spectr. 2025;13(10):e0168425. https://doi.org/10.1128/spectrum.01684-25.

12. Ajoolabady A, Pratico D, Tang D, et al. Immunosenescence and inflammaging: Mechanisms and role in diseases. Ageing Res Rev. 2024;101:102540. https://doi.org/10.1016/j.arr.2024.102540.

13. Liu B, Bao ZY, Chen W, et al. Targeted Next-Generation Sequencing in Pneumonia: Applications in the Detection of Responsible Pathogens, Antimicrobial Resistance, and Virulence. Infect Drug Resist. 2025;18:407-18. https://doi.org/10.2147/IDR.S504392.

14. Hong S, Zhu M, Huang Y, et al. Post-pandemic resurgence of respiratory virus infections: age and department-specific patterns in a Chinese tertiary hospital (2020-2024). BMC Infect Dis. 2025; 25(1):1431. https://doi.org/10.1186/s12879-025-11919-x.

15. Akinosoglou K, Schinas G, Almyroudi MP, et al. The impact of age on intensive care. Ageing Res Rev. 2023;84:101832. https://doi.org/10.1016/j.arr.2022.101832.

16. Ballesteros I, Hidalgo A. The neutrophil collective. Cell. 2025;188(25):7019-35. https://doi.org/10.1016/j.cell.2025.11.001.

17. Torfs K, Vermeersch G, Gouwy M, et al. Neutrophils as critical orchestrators of chronic inflammation. Cell Mol Immunol. 2026;23(2):123-49. https://doi.org/10.1038/s41423-025-01380-w.

18. Khoyratty TE, Ai Z, Ballesteros I, et al. Distinct transcription factor networks control neutrophil-driven inflammation. Nat Immunol. 2021;22(9):1093-1106. https://doi.org/10.1038/s41590-021-00968-4.

19. Poli V, Zanoni I. Neutrophil intrinsic and extrinsic regulation of NETosis in health and disease. Trends Microbiol. 2023;31(3):280-93. https://doi.org/10.1016/j.tim.2022.10.002.

20. Tu H, Ren H, Jiang J, et al. Dying to Defend: Neutrophil Death Pathways and their Implications in Immunity. Adv Sci (Weinh). 2024;11(8):e2306457. https://doi.org/10.1002/advs.202306457.

21. Tian J, Liu C, Wang X, et al. Comparative analysis of clinical features of lower respiratory tract

infection with respiratory syncytial virus and influenza virus in adults: a retrospective study. BMC Pulm Med. 2023;23(1):350. https://doi.org/10.1186/s12890-023-02648-5.

22. Zhang M, Xu N, Cheng Q, et al. Immune status assessment based on plasma proteomics with meta graph convolutional networks. BMC Genomics. 2025;26(1):360. https://doi.org/10.1186/s12864-025-11537-6.

23. Mittelbrunn M, Guido Kroemer G. Hallmarks of T cell aging. Nat Immunol. 2021;22(6):687-98. https://doi.org/10.1038/s41590-021-00927-z.

24. Davies K, McLaren JE. Destabilisation of T cell-dependent humoral immunity in sepsis. Clin Sci (Lond). 2024;138(1):65-85. https://doi.org/10.1042/CS20230517.

25. Starshinova AA, Kudryavtsev I, Malkova A, et al. Molecular and Cellular Mechanisms of M. tuberculosis and SARS-CoV-2 Infections-Unexpected Similarities of Pathogenesis and What to Expect from Co-Infection.Int J Mol Sci. 2022;23(4):2235. https://doi.org/10.3390/ijms23042235.

26. Cumming E, Peters C. Immune response to infection. Anaesthesia & Intensive Care Medicine. 2024;25(6):439-43. https://doi.org/10.1016/j.mpaic.2024.04.003.

27. Ward ES, Gelinas D, Dreesen E, et al. Clinical Significance of Serum Albumin and Implications of FcRn Inhibitor Treatment in IgG-Mediated Autoimmune Disorders. Front Immunol. 2022;13: 892534. https://doi.org/10.3389/fimmu.2022.892534.

28. Pajic-Lijakovic I, Milivojevic M, Barshtein G, et al. The Mechanical Properties of Erythrocytes Are Influenced by the Conformational State of Albumin. Cells. 2025;14(15):1139. https://doi.org/10.3390/cells14151139.

29. Kdimati S, Mullins CS, Linnebacher M. Cancer-Cell-Derived IgG and Its Potential Role in Tumor Development. Int J Mol Sci. 2021; 22(21):11597. https://doi.org/10.3390/ijms222111597.

30. Naddaf M. Long-COVID signatures identified in huge analysis of blood proteins. Nature. 2024; 626(7997):19-20. https://doi.org/10.1038/d41586-024-00158-w.

31. López-Otín C, Blasco MA, Partridge L, et al. Hallmarks of aging: An expanding universe. Cell. 2023;186(2):243-78. https://doi.org/10.1016/j.cell.2022.11.001.

32. Azzini E, Peluso I, Intorre F, et al. Total and Plant Protein Consumption: The Role of Inflammation and Risk of Non-Communicable Disease. Int J Mol Sci. 2022;23(14):8008. https://doi.org/10.3390/ijms23148008.

33. Ren YJ, Yan CK, Yang H. Erythrocytes: Member of the immune system that should not be ignored. Crit Rev Oncol Hematol. 2023;187:104039. https://doi.org/10.1016/j.critrevonc.2023.104039.

34. Nairz M, Weiss G. Iron in infection and immunity. Mol Aspects Med. 2020; 75:100864. https://doi.org/ 10.1016/j.mam.2020.100864.

35. Petitdemange C, Funderburg N, Zaunders J, et al. Editorial: Infectious Agent-Induced Chronic Immune Activation: Causes, Phenotypes, and Consequences. Front Immunol. 2021;12:740556. https://doi.org/10.3389/fimmu.2021.740556.

36. Lalbiaktluangi C, Yadav MK, Singh PK, et al. A cooperativity between virus and bacteria during respiratory infections. Front Microbiol. 2023;14:1279159. https://doi.org/10.3389/fmicb.2023.1279159.

37. Jeannet R, Daix T, Formento R, et al. Severe COVID-19 is associated with deep and sustained multifaceted cellular immunosuppression. Intensive Care Med. 2020;46(9):1769-71. https://doi.org/ 10.1007/s00134-020-06127-x.

38. Aguado J, Amarilla AA, Taherian Fard A, et al. Senolytic therapy alleviates physiological human

brain aging and COVID-19 neuropathology. Nat Aging. 2023;3(12):1561-75. https://doi.org/10.1038/s43587-023-00519-6.

39. Li L, Sun F, Han L, et al. PDLIM2 repression by ROS in alveolar macrophages promotes lung tumorigenesis. JCI Insight. 2021;6(5):e144394. https://doi.org/10.1172/jci.insight.144394.